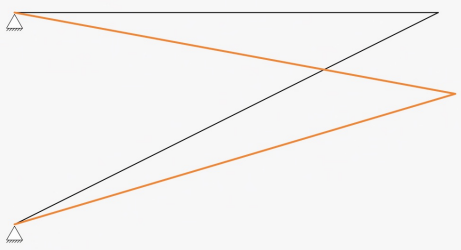
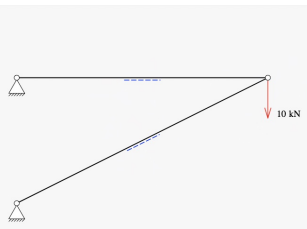
Übungsblatt 1

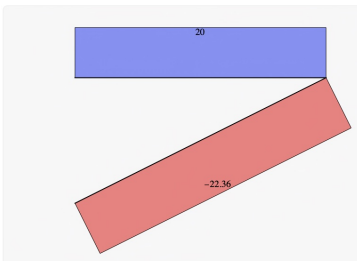
Aufgabe 1

Tragwerk | Zwei Stäbe | Drei Stäbe | Fachwerk

u = {0.016, 0.0767}



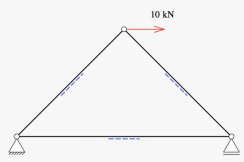


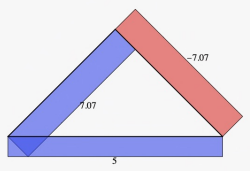


Tragwerk | Zwei Stäbe | Drei Stäbe | Fachwerk

u = {0.255, 0.0667, 0.133}

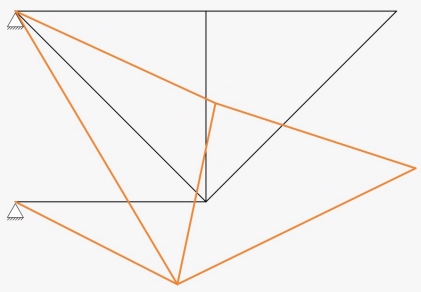


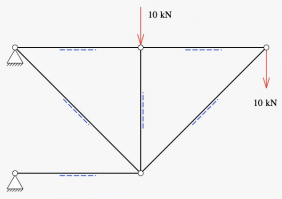


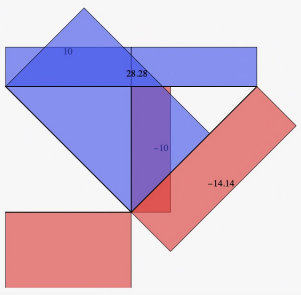


Tragwerk | Zwei Stäbe | Drei Stäbe | Fachwerk

u = {−0.06, 0.173, 0.02, 0.193, 0.04, 0.33}







Aufgabe 2

Vorab

$$A = 51^2 - 49^2 = 200 \text{ mm}^2$$

$$E = 210000 \text{ N/mm}^2$$

$$EA = 210000 \cdot 200 \cdot 10^{-3} = 42000 \text{ kN}$$

$$l_0 = \sqrt{3^2+4^2} = 5 \qquad \underline{x}_1 = \begin{pmatrix} 1 \\ 0 \end{pmatrix}$$

$$\cos\alpha_0 = 3/5 \quad = 0.6 \qquad \underline{x}_2 = \begin{pmatrix} 4 \\ 4 \end{pmatrix}$$

$$\sin\alpha_0 = 4/5 \quad = 0.8$$

2.1 Geometrisch exakt

Verformen $\underline{u}_1, \underline{u}_2$

$$l = \left| \begin{pmatrix} 1 \\ 0 \end{pmatrix} + \begin{pmatrix} 0.1 \\ 0.05 \end{pmatrix} - \left(\begin{pmatrix} 4 \\ 4 \end{pmatrix} + \begin{pmatrix} 0.14 \\ 0.09 \end{pmatrix} \right) \right|$$

$$= 5.056 \text{ m}$$

$$\varepsilon = \frac{5.056 - 5}{5} = 0.0112$$

$$N = 42000 \cdot 0.0112 = 470.5 \text{ kN}$$

$$\underline{d} = \frac{1}{5.056} \cdot \left(\begin{pmatrix} 1 \\ 0 \end{pmatrix} + \begin{pmatrix} 0.1 \\ 0.05 \end{pmatrix} - \left(\begin{pmatrix} 4 \\ 4 \end{pmatrix} + \begin{pmatrix} 0.14 \\ 0.09 \end{pmatrix} \right) \right) = \begin{pmatrix} 0.6013 \\ 0.7990 \end{pmatrix}$$

$$\underline{P}_2 = 470.5 \cdot \begin{pmatrix} 0.6013 \\ 0.7990 \end{pmatrix} = \begin{pmatrix} 282.9 \\ 375.9 \end{pmatrix}$$

Damit

$$\underline{P} = \begin{pmatrix} -282.9 \\ -375.9 \\ 282.9 \\ 375.9 \end{pmatrix}$$

Vektoren $\underline{v}_1, \underline{v}_2$

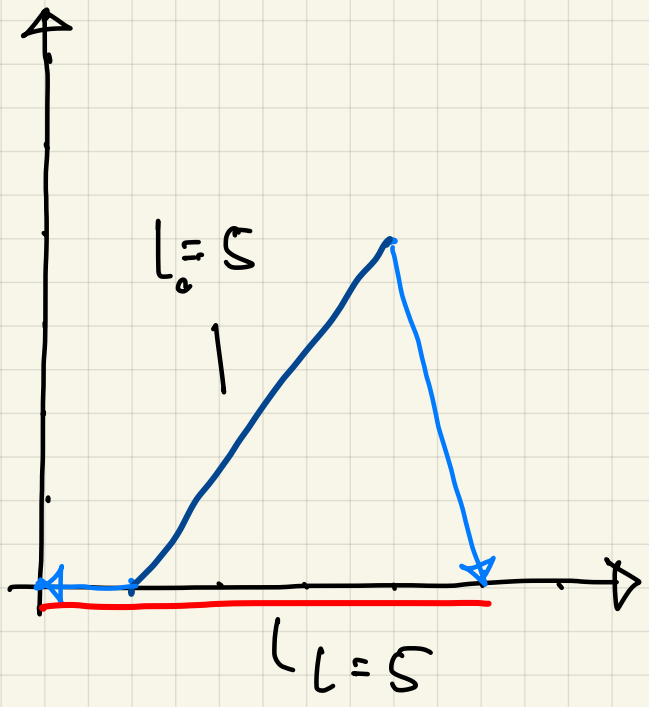
$$L = |\begin{pmatrix}4\\4\end{pmatrix} + \begin{pmatrix}1\\-4\end{pmatrix} - \left(\begin{pmatrix}1\\0\end{pmatrix} + \begin{pmatrix}-1\\0\end{pmatrix}\right)|$$

$$= |\begin{pmatrix}5\\0\end{pmatrix}| = 5$$

$$\varepsilon = \frac{5-5}{5} = 0$$

$$N = EA \cdot 0 = 0$$

$$P = \begin{pmatrix}0\\0\\0\\0\end{pmatrix}$$



Kann das sein?

Ja! Der Stab ändert seine Länge nicht!

2.2 Elementsteifigkeitsmatrix

$$t_0 = \begin{pmatrix} -0.6 \\ -0.8 \\ 0.6 \\ 0.8 \end{pmatrix}$$

$$\underline{K} = \frac{EA}{l_0} \cdot \underline{t}_0 \underline{t}_0^T$$

$$= \frac{42000}{5} \cdot \begin{pmatrix} -0.6 \\ -0.8 \\ 0.6 \\ 0.8 \end{pmatrix} \begin{pmatrix} -0.6 & -0.8 & 0.6 & 0.8 \end{pmatrix}$$

$$= 8400 \cdot \begin{pmatrix} 0.36 & 0.48 & -0.36 & -0.48 \\ 0.48 & 0.64 & -0.48 & -0.64 \\ -0.36 & -0.48 & 0.36 & 0.48 \\ -0.48 & -0.64 & 0.48 & 0.64 \end{pmatrix}$$

2.3 Vereinfachte Stabendkräfte

$\underline{P}^{ex}$ sind die Stabendkräfte nach gem. exakter Formulierung.

Verformen $\underline{u}_1, \underline{u}_2$

$$\underline{P} = \underline{K} \begin{pmatrix} \underline{u}_1 \\ \underline{u}_2 \end{pmatrix} = \begin{pmatrix} -282.2 \\ -376.3 \\ 282.2 \\ 376.3 \end{pmatrix} \qquad \underline{P}^{ex} = \begin{pmatrix} -282.9 \\ -375.9 \\ 282.9 \\ 375.9 \end{pmatrix}$$

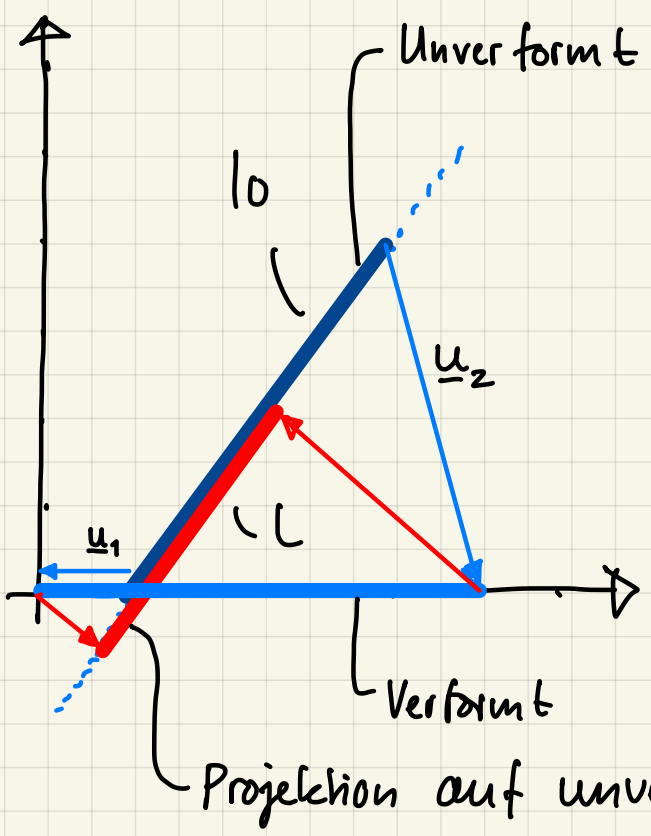
Ok., Unterschied klein!

Verdrehen $\underline{v}_1, \underline{v}_2$

$$\underline{P} = \underline{K} \begin{pmatrix} \underline{v}_1 \\ \underline{v}_2 \end{pmatrix} = \begin{pmatrix} 10080 \\ 13440 \\ -10080 \\ -13440 \end{pmatrix} \qquad \underline{P}^{ex} = \begin{pmatrix} 0 \\ 0 \\ 0 \\ 0 \end{pmatrix}$$

Nicht Ok, Unterschied riesig!

Erklärung



Länge l aus Projektion auf Ausgangslage.

Vereinfachung hier nicht zulässig!

Aufgabe 3

Alternative Formulierung

$$\underline{K} = \underline{T}^T \underline{k}\, \underline{T} \quad \text{mit}$$

$$\underline{k} = \frac{EA}{L_0}\begin{pmatrix} 1 & -1 \\ -1 & 1 \end{pmatrix} \quad \text{und} \quad \underline{T} = \begin{pmatrix} c & s & 0 & 0 \\ 0 & 0 & c & s \end{pmatrix}$$

wobei $c = \cos\alpha_0$, $s = \sin\alpha_0$

$$\underline{K} = \frac{EA}{L_0}\begin{pmatrix} c & 0 \\ s & 0 \\ 0 & c \\ 0 & s \end{pmatrix}\begin{pmatrix} 1 & -1 \\ -1 & 1 \end{pmatrix}\begin{pmatrix} c & s & 0 & 0 \\ 0 & 0 & c & s \end{pmatrix}$$

$$= \frac{EA}{l_0}\begin{pmatrix} c & -c \\ s & -s \\ -c & c \\ -s & s \end{pmatrix}\begin{pmatrix} c & s & 0 & 0 \\ 0 & 0 & c & s \end{pmatrix}$$

$$= \frac{EA}{l_0}\begin{pmatrix} c^2 & cs & -c^2 & -cs \\ cs & s^2 & -cs & -s^2 \\ -c^2 & -cs & c^2 & cs \\ -cs & -s^2 & cs & s^2 \end{pmatrix}$$

→ Selbe Matrix wie im Skript S. 14.